

THERMACART · DESIGN DECISION RECORD

ThermaCart TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, the items that remain open, and the TRL 3 refinements that follow from TCT-CAL-001

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations in items D1 to D8 are adopted for TRL 3 work pending Amish's review; items O1 to O9 remain proposed, awaiting Amish.

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed eight items as "Proposed, awaiting Amish", and the design precis TCT-PRC-001 v0.2 listed six key design choices, each with options and, for most, a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under that instruction and stays open for his review. Items without a recommendation stay "Proposed, awaiting Amish". Nothing here is recorded as decided or approved by Amish.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in TCT-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items adopted for TRL 3 work.

#	Item	Recommendation adopted	Status
D1	Envelope	Option (a): a stock 6 x 2 x 1/8 in aluminium tube sized to a GN 1/3 slot; the WHO PQS 0.6 L envelope (b) kept as a later small size, a custom finned extrusion (c) only at volume	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D2	Grades	C5, C25 and H70 in organic paraffin; water (W0) as a produce-only option; salt hydrates deferred to a lined variant; C5 is the first build	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D3	Pitch wording	Option (b): "cold chain, comfort and hot holding" instead of "cold chain, comfort and cooking", because the H70 grade holds cooked food hot and does not cook. Applied to <code>project.yaml</code> and <code>README.md</code>	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D4	Grade keying and indicator	Mechanical nose key per grade, key table published as part of the open standard; passive sight window over a vial of the same PCM rather than a thermochromic strip	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D5	Closing the C5 gaps	Evaluate a 2 mm wall at TRL 3. Done in TCT-CAL-001: the 2 mm wall adds 3.5 % volume and would reach 1.28 times the minimum yield under the vacuum of a frozen cartridge, so it is not adopted; with verified datasheet values C5 meets R3 (65.4 Wh) on the 3.175 mm tube	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D6	Budget	Option (a): keep <code>budget_usd</code> at \$80 for one C5 cartridge and one frame; the three-grade set is outside the budget. No new budget figure was recommended, so <code>budget_usd</code> is unchanged. TCT-CAL-001 prices that build at \$87, \$7 over (see O6)	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D7	First host type	A produce cooler user, reached through a local partner, because R10 is the tightest requirement. Only the host type is adopted; no partner is named	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D8	Fill and shell	Fill hot and seal for life (now: fill as liquid at each grade's maximum operating temperature); aluminium shell for paraffins, no HDPE	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review

Table 2. TRL 3 refinements made within D1, D4, D6 and D8 because of TCT-CAL-001, open for Amish's review.

#	Refinement	Reason
T1	Folding bail handle on phenolic thermal-break washers, stowed within the GN 1/3 length, in place of the fixed handle	The fixed handle left about 14 mm of finger room; the bail gives 39 mm and keeps R2 (323.4 mm)
T2	Caps built as a stack of 3.175 mm flange, 3 mm gland spacer and 9.525 mm plug, with a spliced FKM cord O-ring in the corner gland and six radial M5 screws; tube lengthened from 270 to 280 mm	Garage-buildable without milling (R16); the thinner flange frees length for PCM
T3	Fins from stock 6.35 x 1.59 mm (1/4 x 1/16 in) 6063 flat bar	Stock section; overall height 63.5 mm, within 65 mm
T4	R12 target corrected from an oven "at 100 °C or less" to "at 85 °C, never above 90 °C"; storage limits of 45 °C (C5) and 65 °C (C25) added to the safety text	Rubitherm datasheet maximum operating temperatures

Table 3. Items that remain open.

#	Item	Status
01	The named co-design partner and region for the first produce cooler host. No recommendation was made beyond the host type in D7	Proposed, awaiting Amish
02	Whether ColdPod or ZeerBox should declare ThermaCart as a dependency. No recommendation was made; neither repo is in this batch and neither was changed	Proposed, awaiting Amish
03	C5 charging: freezer with a conditioning step, or a refrigerator at 0 to 2 °C. No recommendation was made at TRL 2; TCT-CAL-001 shows refrigerator charging takes 31 to 51 h	Proposed, awaiting Amish
04	R5 (PCM fraction 38 to 40 % against 50 %): relax the target for the stock-tube build, or keep it for a later extrusion	Proposed, awaiting Amish (see docs/REVIEW.md)
05	R10 (cold hold about 2 h against 8 h): dark finish, contact with the load, more cartridges, or a revised requirement	Proposed, awaiting Amish (see docs/REVIEW.md)
06	R16 and budget: \$71 per cartridge against \$50, and \$87 for the D6 build against \$80	Proposed, awaiting Amish (see docs/REVIEW.md)
07	R14: handling rule for an H70 cartridge taken straight from an oven	Proposed, awaiting Amish (see docs/REVIEW.md)
08	Surface finish of the shell (mill finish or dark paint or anodizing), which affects R10, R11 and R12	Proposed, awaiting Amish (see docs/REVIEW.md)
09	R1 for C25: widen the C25 melt band to 22 to 27 °C, or look for a paraffin that melts within 24 to 27 °C	Proposed, awaiting Amish (see docs/REVIEW.md)

■ Consequences

- `project.yaml` : pitch reworded under D3; TRL fields set to 3. `budget_usd` stays at \$80 and `problem` is unchanged.

- TCT-PRB-001, TCT-PRC-001 and TCT-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed"; they carry the D1 to D8 wording. Requirement changes: R12's oven temperature (T4) and a note under R16 that the \$80 budget covers one C5 cartridge and one frame (D6). No other target is relaxed.
- TCT-CAL-001 finds R5, R10 and R16 not met and R1 (C25 melt band), R12 and R14 at risk. Options for each are in `docs/REVIEW.md` and await Amish; this record does not decide them.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.